CrackW-Net: A Novel Pavement Crack Image Segmentation Convolutional Neural Network

Chengjia Han, Tao Ma, Ju Huyan, Xiaoming Huang, and Yanning Zhang

***Abstract*— Image-based intelligent detection of road cracks with high accuracy and efficiency is vital to the overall condition assessment of the pavement. However, significant problems of continuous cracks interruption and background discrete noise misidentification are frequently observed in current semantic segmentation of pavement cracks, which mainly caused by traditional segmentation convolutional neural networks. This paper proposes a skip-level round-trip sampling block structure with the implementation of convolutional neural networks, thereby constructed a novel pixel level semantic segmentation network called CrackW-Net. After that, two datasets, including the widely recognized Crack500 dataset and a self-built dataset, were used to train two versions CrackW-Net, FCN, U-Net and ResU-Net. Meanwhile, comparative experiments are conducted among all these network models for crack detection. Results show that CrackW-Net without residual block performs the best in the task of pavement crack segmentation.**

***Index Terms*— Convolutional neural network, pavement crack, pavement detection, road maintenance, semantic segmentation.**

I. Introduction

With continues increasing of the demand for road maintenance, computer vision based evaluation of road conditions has become a research hotspot in the industry [1]–[3]. The research of applying computer vision technologies for pavement damage condition assessment has been developed for decades. In general, there are two branches of methodologies in this research field. The first branch is based on the traditional image processing techniques [4]. This branch of methodologies is represented by the development of various filters such as edge detectors, which is based solely on the characteristics of the optical images themselves [5]. Despite the extensive research of these methods, the shortcomings of them are obvious. It is usually very difficult for these methods to detect irregular targets under complex background environments and illumination conditions, because the methods cannot understand the meaning of the contents in the images [6].

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The authors are with the Department of Road Engineering, Southeast University, Nanjing 211189, China (e-mail: hanchengjia@seu.edu.cn; matao@seu.edu.cn; ju_huyan@outlook.com; huangxm@seu.edu.cn; 38416993@qq.com).

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The second branch of methods is based on the latest artificial intelligence technologies, i.e., deep learning. This branch has attracted significant attention in recent years due to its robustness in complex environments [7]. The core of deep learning-based computer vision techniques is the convolutional neural network (CNN) architecture. It has been well accepted that CNN can achieve significantly higher generalization and detection accuracy compared with traditional image processing algorithms. The prototype "neocognitron" of this structure was first proposed by Japanese scientist Kunihiko Fukushima in 1980. However, limited by computing power, less attention was given in the following 20 years [8]. In 2006, researchers successfully used graphics processing unit (GPU) to accelerate CNN, which made the training of deep neural networks possible [9]. Until 2012, the emergence of "AlexNet" marked the resurgence of neural networks and the rise of deep learning. It reached the top with an error rate of 15.3% on the image classification task of ImageNet2012 [10]. Since then, a growing number of scholars in road engineering have tried to improve the CNN architecture and use it for inspection tasks such as road cracks detection. At the end of 2015, Kaiming He made another breakthrough of proposing "ResNet", which brought the residual structure to the original neural network model with brilliant performance enhancement [11]. In 2017, "DenseNet" network was proposed to further improve the computing performance of the network. After that, the convolutional neural network began to optimize based on different task requirements [12].

In the direction of target detection and image segmentation, there are typically two branches of methods based on deep convolutional networks. One method is to process the target detection, segmentation task and classification one by one. Methods that have adopted this idea include Faster-RCNN and Mask-RCNN [13]. The other is the pixel-level segmentation method, which can perform three types of tasks in a specific order in the network at the same time, and these tasks are highly coupled for training. Typical methods in this category include Fully Convolutional Networks (FCN) [14] and U-Net [10].

Many scholars applied these networks to the problem of road surface damage recognition and optimize the network or loss function according to the characteristics of these image dataset. Zhang *et al.* [6] proposed an efficient architecture CrackNet based on the Convolutional Neural Network for pixel-perfect automated pavement crack detection on 3D asphalt. Ni *et al.* [15] proposed a convolutional

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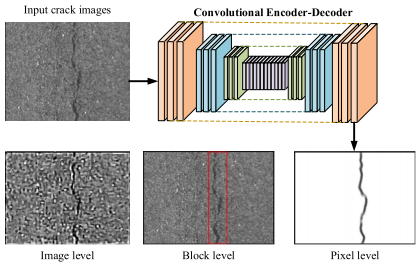

Fig. 1. Crack detection trend with different analysis levels [19].

neural network-based framework that achieved pixel-level crack extractions with high accuracy and efficiency through convolutional neural network, which also achieve pixel-level crack classifications. Park *et al.* [16] proposed a convolutional neural network composed of segmentation and classification modules to process road crack images taken by vehicles. Mei *et al.* [17] proposed a novel loss function for automatic pavement crack detection, which considered the connectivity of pixels to overcome the issues related to the output of transposed convolution layers. After that, they proposed a novel deep-learning architecture for crack segmentation at pixel-level based on multi-level feature fusion [18]. Ju *et al.* [19] proposed a state-of-the-art pixelwise crack detection architecture called CrackU-net, which is featured by its utilization of advanced deep convolutional neural network technology.

Based on these studies, this paper proposes a new pixel-level pavement crack segmentation network structure, CrackW-Net. This new net is tested and compared with mainstream pixel-level segmentation networks on the same road crack dataset. The results show that CrackW-Net has excellent segmentation accuracy and performs well in the validation set, which means that this network has a strong generalization ability. Specifically, this paper is organized as follows: Section 2 summarizes the state of the art technologies used in pixel-level crack detection models. Section 3 explains the structure of CrackW-Net. Section 4 conducted the model training for CrackW-Net and the others and compares it with other networks. Section 5 discusses findings from the experimental results. Section 6 concludes the overall contributions of this paper.

II. Overview of Common Components of Pixel-Level Segmentation Network

Image based road crack detections can generally be divided into image-level, block-level and pixel-level, as shown in Figure 1. This article focuses on the pixel-level image segmentation based on the convolutional neural network. The main principle is to extract high-dimensional feature information from the input image through convolution operations, and then restore the feature size through transposed convolution, by which to achieve pixel level crack segmentation from the output. Since the predictions of crack or non-crack are conducted on each pixel values on the image, so this type of crack detection is usually referred as pixel-level segmentation. This section summarized the general components of a typical pixel-level convolutional neural network.

A. Convolution

Convolutional layer is the most important part of a deep image segmentation neural network. It was first proposed by LeCun *et al.* [8]. Convolution operation is a specific mathematical operation that slides the filter on the image while doing weighted sums between the inputs and weights of the specific neural nodes at the corresponding positions. The outputs should include the bias if those values exist. Based on this principle, the output of a network layer can be calculated by Equation 1 [17].

$$O(i, j.k) = \sum_m \sum_n \sum_l I(i-m, j-n, l) w_k(m, n, l) + b_k \quad (1)$$

where I and O stand for the input and output features of the convolutional layer. l and k represent the *l-th* and *k-th* features of the input and output.

From the perspective of information transformation, convolution process is a process of transforming and mapping the information of original image into higher dimension that removes the dependent information while remains independent and informative information, which are achieved through non-linearity transformation of the activation function. Moreover, the design of convolutional neural network can share weights, which greatly reduces the amount of parameters compared with full connection.

B. MaxPooling and AveragePooling

Pooling operation is a special form of convolution operation. Instead of multiplying and summing the filter and the corresponding position of the image [20], pooling operation directly takes the largest value (corresponding to MaxPooling) or average value (corresponding to AveragePooling) in the filter area as output, which are defined in Equation 2 and 3 [17]. The primary objective of pooling operation is to reduce the number of parameters while maintain the key features, which also called down-sampling by some literature. Therefore, the size of feature maps is significantly reduced after pooling operations, which can reduce the number of parameters and the probability of overfitting simultaneously.

$$O_{Max\,Pooling}(i, j, k) = \underset{p=-t}{\overset{t}{Max}} \underset{q=-t}{\overset{t}{Max}} (I(i-p, j-q, l)) \quad (2)$$

$$O_{Average\,Pooling}(i, j, k) = \underset{p=-t}{\overset{t}{Mean}} \underset{q=-t}{\overset{t}{Mean}} (I(i-p, j-q, l)) \quad (3)$$

where i and j are indices of the input features, x and y are indices of the output and $2t+1$ is the filter size.

C. Transposed Convolution

Transpose convolution is an inverse operation of convolution operation. Intuitively, transpose convolution can increase the image size. Transpose convolution operation is achieved by first filling the recorder with 0 and rearranges it according to the size of the original image and the step size of the recorder, then transposing the matrix and multiply it with the original image. Finally, rearranges the result into the output image

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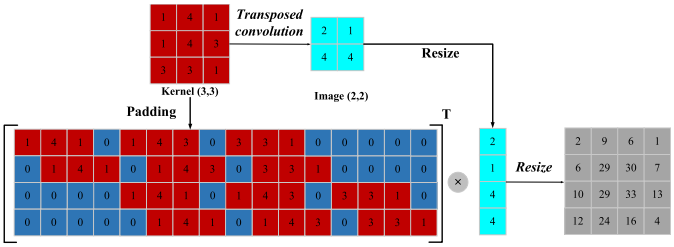

Fig. 2. Examples of transposed convolution[22].

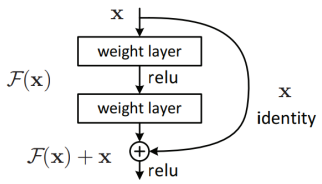

Fig. 3. Residual learning: a building block [11].

of the current size [21]. Transposed convolution is the key operation in realizing pixel-level image segmentations. The process of transposed convolution is shown in Figure 2.

D. Residual Block

Residual block is first proposed and applied to ResNet [11] to solve the degradation problem of deep convolution network. When the network reaches a certain depth, the performance of the model will temporarily fall into a bottleneck that is difficult to increase. When the network continues to deepen, the performance of the model on the test set will decrease. This phenomenon is called network degradation. He *et al.* believes that the reason for the degradation is the difficulty of directly fitting identity mappings $H(x) = x$, which exist in abundance in the network deep layers. However, if the function is designed as $H(x)=F(x)+x$, as shown in Figure 3, then, when $F(x)=0$, $H(x)$ can be regarded as an identity mapping. The difficulty of fitting the residual function ($F(x)=H(x)-x$) thus will be greatly reduced.

The residual network has become the popular configuration of deep convolutional networks, and we also applied this block to the proposed CrackW-Net in this article.

E. U-NET Framework

U-Net is a classical convolutional network for semantic segmentation, which uses multi-scale features for semantic segmentation [23]. The name of the network based on its U-shape, as Figure 4 shows. Firstly, the input mage is compressed by a series of down-sampling operations composed of convolution and Max pooling (called *Contracting path* in paper), and then the original size of the image is restored by a series of up-sampling operations composed of transposed convolution and concatenate operations (called *Expanding path* in paper).

U-Net has excellent segmentation performance for small sample datasets. However, the original version of U-Net has some problems, such as the disorder of image size caused by

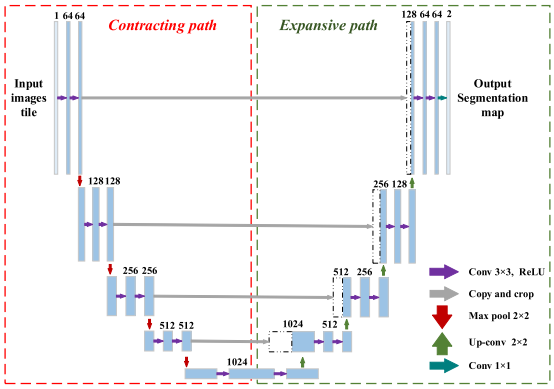

Fig. 4. U-Net framework [10].

not using convolution layer of "same" mode, which reduces the universality of the network. In this paper, a new pixel-level segmentation network—CrackW-Net is designed based on the general structure of U-Net.

III. CrackW-Net Architecture

This research proposes a pixel-level convolutional neural network for road crack segmentation, which is called CrackW-Net. CrackW-Net is based on U-Net, while improves the pooling and up-sampling methods considering the special characteristic and the difficulties of pavement crack segmentation. The main improvement is adding a skip-level round-trip sampling block, which does not involve updating of the convolution operation, so it is easy to build and test.

When down and up sampling through a convolutional network, the receptive field of down-sampling and up-sampling is fixed (2×2) or (3×3) in traditional networks. Sampling with a fixed sampling scale will make the network scale-dependent, which is not conducive to the distinguishing of image features of different scales. This problem is very prominent during pavement crack segmentation. A complete crack is likely to be interrupted during segmentation, or the proportion of independent spots in the recognition map increases significantly when there are many pits on the road surface. To some extent, these problems are due to the limitation of the sampling scale, which makes the network recognition of features close to a certain small scale and loses the global receptive field.

In order to improve the ability of different receptive fields to perceive information at different scales, the idea of DenseNet and add skip-level feature fusion were adopted in the proposed network [24]. The difference is that in order to reduce the network size and training cost, instead of using dense connection block to replace the existing convolution layers, we propose a skip-level round trip sampling block to combining the characteristics of different receptive fields. We believe that the fusion of multi-scale pooling (or up-sampling) can expand the network receptive field to improve the ability to distinguish between background, interference information and segmented objects, as shown in Figure 5.

Specifically, we used the U-Net as a prototype. After the input images is down-sampled to the smallest feature map by (2×2), the up-sampling and concatenate operations are

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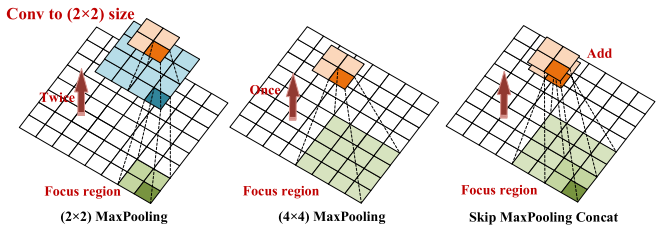

Fig. 5. Feature receptive field expansion based on multi-scale pooling fusion.

not directly performed to realize the pixel prediction. Instead, up-sampling is performed with a receptive field (4×4), which is twice as large as the initial down-sampling, and the feature map in the original down-sampling is copied and concatenated by skipping the level. When we return the feature map to the original image size by twice up-sampling, we expand the receptive field again to (16×16), which is 4 times as large as the initial down-sampling, and directly down-sample the original sampling size to restore the image to the smallest size feature map set. After that, we merge the new feature map obtained by up and down sampling with the feature map that was originally obtained through small-size down-sampling, followed by the operation of U-Net to restore the original size of the image. Specifically, 5 times (2×2) size up-sampling is performed and after each up-sampling, it is spliced with the corresponding scale down-sampling feature images. Finally, the predicted pixel values are output pixel by pixel. Since the shape of the additional skip level round trip sampling block is an inverted "V", we named this new network as CrackW-Net, as shown in Figure 6.

The other reason why we added up and down sampling actions is to enhance the network's memory of transmitting low-level features in the shallow layer. When the features are extracted by the network step by step, the features of the cracks are repeatedly fitted and emphasized, but excessive emphasis on the crack features will make the network very sensitive to changes in road pixels, which leads to pitting and other non-crack features being treated as cracks deal with. Therefore, we use round-trip sampling actions to directly parallel low-level features with high-dimensional sampling features. The purpose is to achieve efficient low-level feature reuse to help the network pay attention to the distinction between background and cracks.

Figure 6 is the detailed structure of CrackW-Net. The lower right corner of the figure explains the types of layers and arrows in the figure. It should be noted that the illustrated CrackW-Net structure is the network structure used in the following case study section. The input image size in the figure is related to the image size of the database we use, and the depth of sampling and the number of convolution kernels in the convolutional layer are related to the GPU performance we can use. The modified parameters in the figure are for reference only and are not the most suitable parameter selection for CrackW-Net network.

IV. Case Study

In this part, *Crack500* dataset [25] is used to test the performance of the network. The *Crack500* dataset is an open-source dataset which is published by a project on *Github* (*https://github.com/fyangneil/pavement-crack-detection*). The pictures in this dataset are crack pictures of cement concrete pavement and the size of the pictures is mostly ($1440 \times 2560 \times 3$). Since the number of open-source images of *Crack500* database is relatively small, we have introduced a self-built pavement crack database (Including cement and asphalt concrete pavement cracks), and further operation such as rotating and turning over the pictures in the *Crack500* dataset and the self-built dataset to expand the training data. Since the image size of the self-built database is ($720 \times 1280 \times 3$), we have also reduced the image in the *Crack500* database to the same size to ensure data quality while increasing the calculation speed of the networks. When we reduce the image, we call the *resize* function in *Opencv* library and use the default bilinear interpolation algorithm. In the end, we created 1000 crack images with size ($720 \times 1280 \times 3$) and corresponding segmentation labels with size ($720 \times 1280 \times 1$). Among them, 100 images are split as the test set, 200 images are used as the verification set, and 700 images are used as the training set.

A. Model Test

For the crack segmentation problem, we designed two versions of CrackW-Net network. The first CrackW-Net type sets the number of convolution kernels to 2 in the final output layer and sets the activation function to "Softmax" (calculated by Equation 4) [26], and outputs segmented images of ($720 \times 180 \times 2$) size.

$$S_i = \frac{e^{V_i}}{\sum_j e^{V_j}} \tag{4}$$

where, V is the element value in the array; j is the number of elements.

We design the target task as a binary classification task and use "Sparse categorical crossentropy" to do one-hot encoding in the loss function calculation process, so that there is no need to change the original segmentation labels [27]. The calculation of the cross-entropy loss function is as follows.

$$Loss = -\frac{1}{n}\sum_n [y \ln a + (1-y)\ln(1-a)] \tag{5}$$

where, n is the number of pixels in the picture; y is the value of the pixel, a is the probability of the pixel.

In the second version, we treat crack segmentation as a regression task to directly predict pixel by pixel. Theoretically, pixel-by-pixel prediction can better test the robustness of the model, especially when facing smaller dataset, so we choose the second version for subsequent analysis. In this case, the number of convolution kernels in the last layer is 1, and we use "Sigmoid" (calculated by Equation 6) as the activation function to limit the pixel value to (0,1) to speed up the convergence of the network.

$$S(x) = \frac{1}{1+e^{-x}} \tag{6}$$

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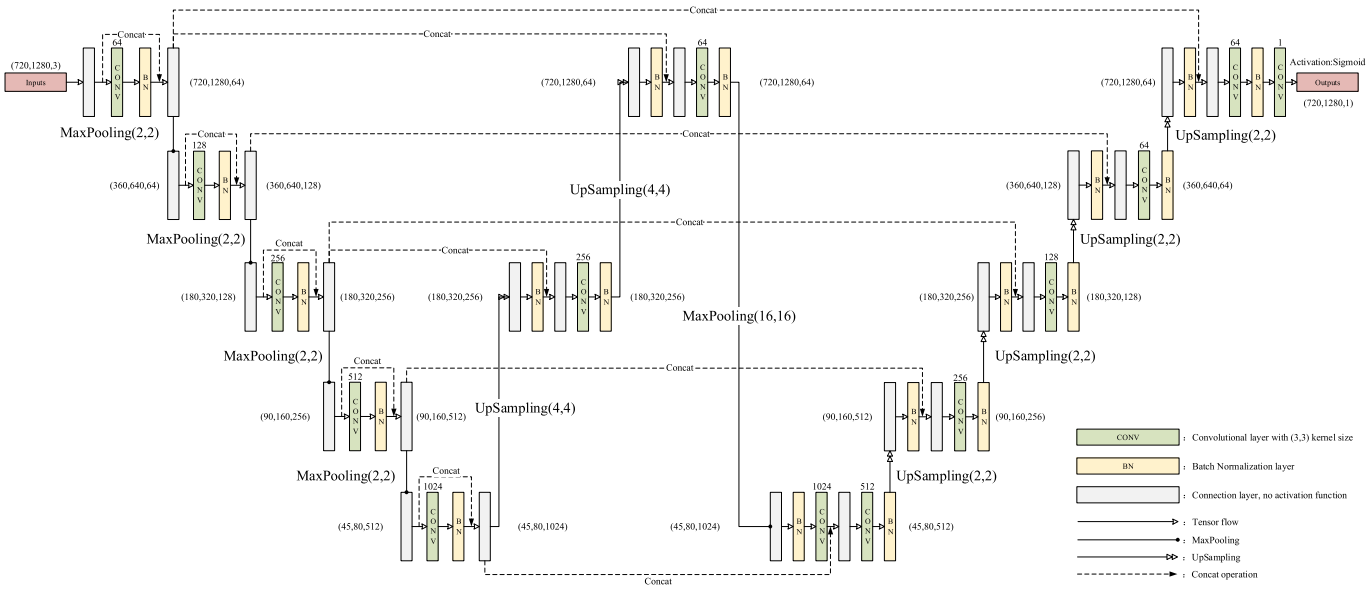

Fig. 6. CrackW-Net framework.

Moreover, we normalized the original segmented label pixels to match it. In the second version, we used "Huber" as the loss function as shown below [28].

$$Loss(y, f(x)) = \begin{cases} \frac{1}{2}(y - f(x))^2 & for \ \ |y - f(x)| \le \delta \\ \delta |y - f(x)| - \frac{1}{2}\delta^2 & otherwise \end{cases} \tag{7}$$

where, δ is the threshold, generally 1; $f(x)$ is the predicted pixel value; y is the true pixel value.

We use the *Tensorflow2.0* [29] computing framework for the realization of CrackW-Net(ResU). In order to compare with CrackW-Net(ResU), we selected three mainstream segmentation networks: FCN, U-Net and ResU-Net. The basic convolutional layer structure of the four networks is exactly the same to ensure that the test results are comparable. U-Net and FCN use this basic convolution structure. The main difference is that U-Net used concatenate operation while FCN used add operation in the multi-scale feature fusion procedure; ResU-Net adds residual structure to each level of the basic convolution structure. CrackW-Net(ResU) add a skip level round trip sampling block (the inverted V structure is in Figure 6) on the basis of ResU-Net. In order to control the number of CrackW-Net parameters at the same level as the control network, we also set the fifth network: CrackW-Net(U) on the basis of U-Net and halve the number of neurons in each layer (For example, convolution layers of 512 kernels are simplified to 256).

In order to make the test results of the four networks comparable, we use the same network optimizer, evaluation index of performance, and number of training epochs. The specific parameters of network training are shown in Table I. Our training GPU is "NVIDIA GeForce RTX 3090 (with 24GB RAM)" and CPU is "Intel(R) Core(TM) i9-9900K CPU @ 3.60GHz (with 64GB RAM)" [31]–[32].

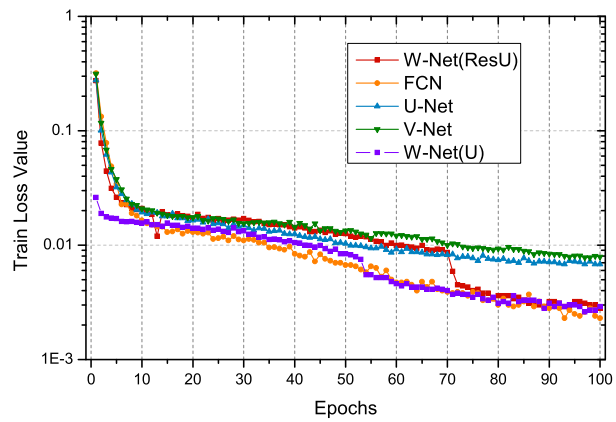

Fig. 7. Comparison of network loss on training set.

After 100 epochs, the training accuracy of the five networks in the training set has achieved convergence. In these Figures, the red line represents CrackW-Net(ResU), the violet line represents Crack W-Net(U), the orange line represents FCN, the blue line represents U-Net, and the green line represents ResU-Net. Figures 7 to 12 are comparisons of the performance parameters of the five networks.

Among them, Figures 7 and 8 are the loss curves of the five networks in the training set and the verification set, respectively. It should be noted that, due to the large change of the network loss value, the ordinates of Figures 7 and 8 are logarithmic. Obviously, the training set loss of FCN, CrackW-Net(ResU) and CrackW-Net(U) after 100 epochs is much smaller than that of the other two networks, while the loss of FCN is slightly smaller than that of CrackW-Net(U) (about 0.0003) and CrackW-Net(ResU) (about 0.0005). However, in the loss of the validation set, CrackW-Net(U) and CrackW-Net(ResU) achieve a clear advantage over the other three networks.

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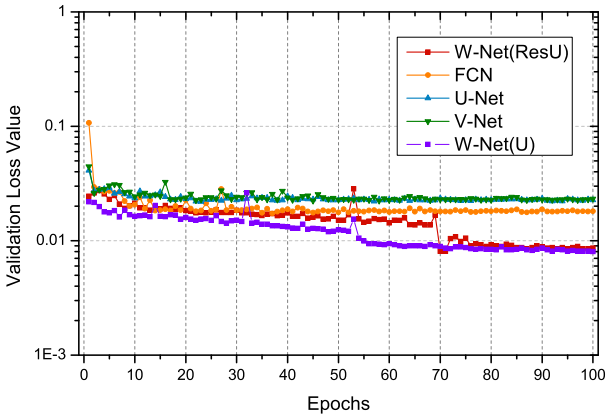

Fig. 8. Comparison of network loss on validation set.

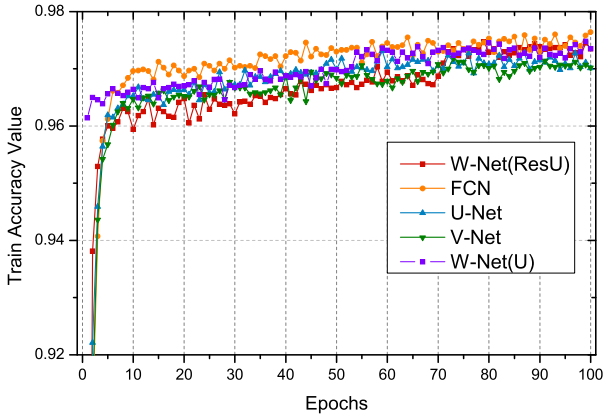

Fig. 9. Comparison of network accuracy on training set.

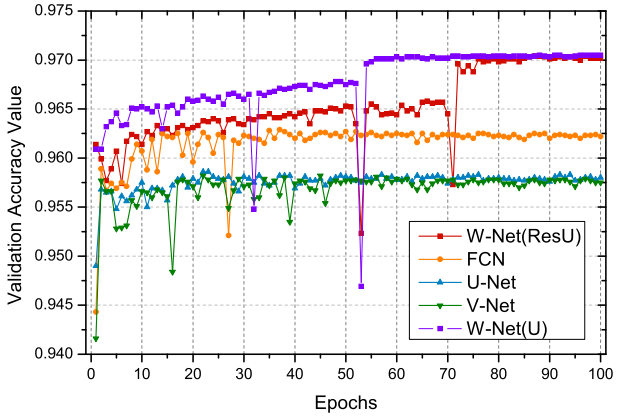

Fig. 10. Comparison of network accuracy on validation set.

Figures 9 and 10 are the accuracy curves of the five networks in the training set and the verification set, respectively. The calculation formula of accuracy is as shown in Equation 8.

$$ACC = \frac{N_{TP} + N_{TN}}{N_{TP} + N_{TN} + N_{FP} + N_{FN}} \tag{8}$$

where, N_{TP} is the number of positive samples classified as positive, N_{TN} is the number of negative samples classified as negative, N_{FP} is the number of negative samples classified as positive, and N_{FN} is the number of positive samples classified as negative.

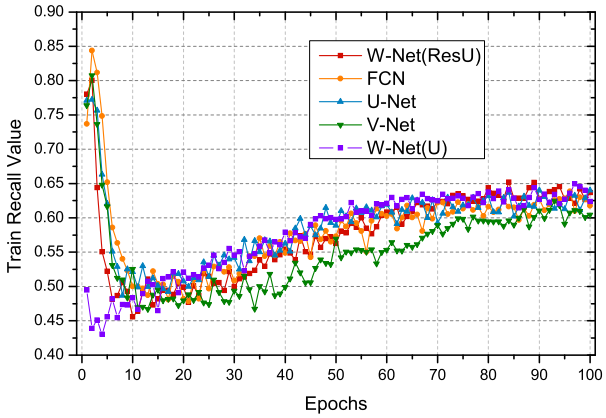

Fig. 11. Comparison of network recall rate on training set.

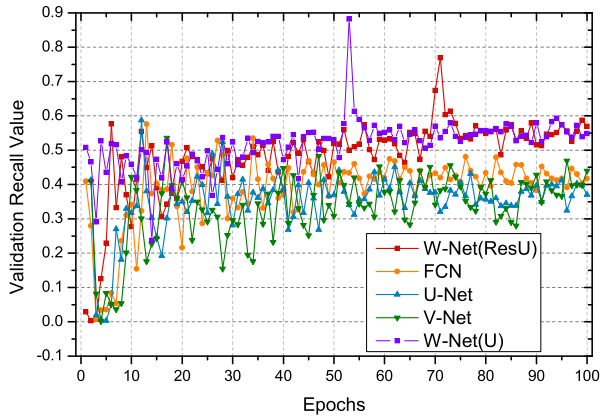

Fig. 12. Comparison of network recall rate on validation set.

TABLE I
PARAMETERS FOR NETWORKS

Loss function	Optimizer	Learning rate
Huber	Adam[30]	1e-3
Metrics	**Batch size**	**Epochs**
[Accuracy, Recall]	8	100

TABLE II
PARAMETERS FOR NETWORKS

Networks	Train Loss	Train ACC	Train Recall	Valid Loss	Valid ACC	Valid Recall	IOU
CrackW(ResU)	0.0028	0.9735	0.6369	0.0086	0.9702	0.5693	0.7104
CrackW(U)	0.0026	0.9742	**0.6492**	**0.0080**	**0.9705**	**0.5737**	**0.7722**
FCN	**0.0023**	**0.9764**	0.6176	0.0181	0.9622	0.4170	0.6954
U-Net	0.0068	0.9701	0.6401	0.0226	0.9580	0.4025	0.6852
ResU	0.0080	0.9707	0.6011	0.0229	0.9575	0.4400	0.6622

It can be seen from Figure 9 and 10 that the accuracy of the five networks in the training set is very close, but in the verification set, CrackW-Net(ResU) and Crack W-Net(U) are

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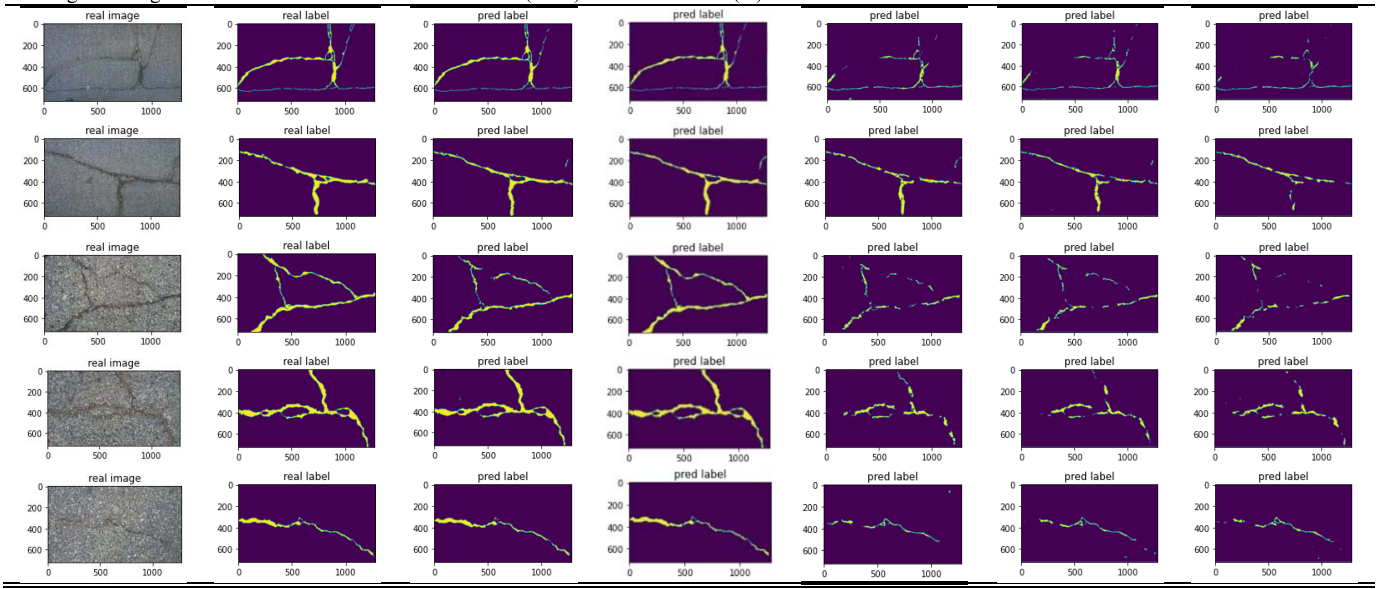
TABLE III

COMPARISON OF THE RESULTS OF THE FIVE NETWORKS ON THE TEST SET

Original image	Ground Truth	CrackW-Net(ResU)	CrackW-Net(U)	FCN	U-Net	ResU-Net
						

much higher than the other three networks. FCN ranks third and U-Net and ResU-Net perform almost the same.

Figures 11 and 12 are the recall curves of the five networks in the training set and the verification set, respectively. The calculation formula of recall is as shown in Equation 9.

$$Recall = \frac{N_{TP}}{N_{TP} + N_{FN}} \tag{9}$$

The performance of the five networks in recall is similar to accuracy. CrackW-Net(ResU) and CrackW-Net(U) have a small lead in the training set, but they have an obvious advantage in the verification set. It should be noted that since the network task is set to regression rather than binary classification, the calculated recall value will be relatively low. The significance of recall value calculation here is to measure the performance difference of network from another perspective, focusing on the relative value rather than the absolute value of recall.

Whether it is CrackW-Net(ResU) or CrackW-Net(U), there will be big or small turns in the ROC curve in the training process, such as the 50th epoch and 70th epoch of the loss curve. Due to the existence of skip-level round-trip sampling block in CrackW-Net, the gradient change is not smooth. Network performance will show a sudden improvement in the training process.

A summary of the final performance indicators of the five networks are shown in Table II, with the best performed value shown in bold. In general, even though two indexes of CrackW-Net(U) in the training set are not the best, they all rank second and have little difference with the best value. The loss is 0.0003 lower than the best and the accuracy is 0.0022 lower than the best. Meanwhile, CrackW-Net(U) has a good performance in all other indicators and far more than the other four networks. The performance of CrackW-Net(ResU) is only slightly lower than that of CrackW-Net(U). Finally, we used the trained five networks to identify cracks on the whole dataset and calculated the IOU(Intersection over Union) value, which reflects the similarity between the prediction region and the ground scene region of the objects in the group of images. The calculation formula of IOU is as shown in Equation 10.

$$IOU = \frac{N_{TP}}{N_{FP} + N_{TP} + N_{FN}} \tag{10}$$

CrackW-Net(U) achieves the largest IOU value, which is much higher than the other four networks, which can be concluded that CrackW-Net has surpassed FCN, U-Net and ResU-Net in pavement crack detections.

After that, we used the extra 100 images to test the performance of the trained five networks for crack segmentations We selected 5 images with the most obvious contrast for display which are shown in Table III. The first two columns of the table are the original image and the manually calibrated ground truth. The last five columns are the segmentation results of CrackW-Net(ResU), CrackW-Net(U), FCN, U-Net and ResU-Net. It can be seen that the crack segmentation accuracies of two kinds of CrackW-Net are obviously promoted, and the discontinuity problem in crack segmentation has been greatly improved. Meanwhile, CrackW-Net(U) is slightly better than CrackW-Net(ResU).

B. Model Paramedic Discussion

This section discussed some engineering issues, which are the number parameters and size of the CrackW-Net model. As mentioned in section 3, we learned from the skip-level link idea of DenseNet but considering the number of parameters and training costs, we removed the dense connection of the convolutional layer. Moreover, it should be noted that the additional skip-level round-trip sampling block used in

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TABLE IV
COMPARISON OF TRAINING COSTS OF THE FIVE NETWORKS

Networks	Total parameters	Non-trainable parameters	Times/step	Model size
CrackW(ResU)	14970465	5888	330ms	180MB
CrackW(U)	4340289	3776	287ms	52.4MB
FCN	3057057	3904	119ms	36.9MB
U-Net	3840417	3904	137ms	46.3MB
ResU	6309249	3904	166ms	76MB

CrackW-Net inevitably increases the number of training parameters. We calculated the parameters of these five networks and the size of the model after training (Save the full model instead of weights), which is shown in Table IV.

The parameter quantity of CrackW-Net(ResU) is twice of ResU-Net and four times of FCN and U-Net. In terms of training speed, each step of CrackW-Net takes twice as much time as the other three networks. However, the parameter quantity of CrackW-Net(U) is almost the same as that of FCN and u-net, which is 1 / 3 smaller than that of ResU-Net. It proves that the skip-level round-trip sampling block proposed in this paper improves the segmentation performance rather than the increase of training parameters.

From the perspective of network size, there are many derivative networks from U-Net network, most of those are improved by intensive skip-level splicing. However, it is skeptical of this practice. In the process of testing, we found that for the small-scale dataset with less categories, the extensive use of skip-level splicing will reduce the training performance of the network. For example, the test effect of ResU-Net in case study is not as good as its original version U-Net, and the test effect of CrackW-Net(ResU) in case study is not as good as CrackW-Net(U). Although the parameter quantity has been increased several times, its performance has not been greatly improved or even decreased. Only from the point of view of this case study, the residual element seems unnecessary. The design of the pixel-level network must be based on the characteristics of the dataset and stacking new technologies sometimes does not necessarily get good results.

Is CrackW-Net(ResU) really worthless? This paper holds that it is not so. The increase of model parameters and residual block are undoubtedly conducive to the improvement of crack network performance. When dealing with larger scale tasks and have to use deeper networks, the increase of training parameters may be acceptable considering its excellent performance. Meanwhile, the negative impact of CrackW-Net(ResU) parameters and training speed can be improved by network optimizations. Due to the structure improvement of CrackW-Net is relatively simple and is not change the convolution operation, the mainstream lightweight methods can be directly used on CrackW-Net, such as Squeeze Net [33], Mobile Net [34], Shuffle Net [35], and Xception [36] etc.

V. CONCLUSION

This paper presents a novel pavement crack image segmentation network named CrackW-Net. The conclusions drawn from this study are as follows:

(1) In order to deal with the problem that continuous cracks are interrupted in segmentation and improve the ability of the network to distinguish the noise points, this paper proposes a skip-level sampling method, and designs the skip-level round-trip sampling block, which can be easily used in various network structures.

(2) Using U-Net as the prototype, this paper designs CrackW-Net with the innovation of the proposed skip-level round-trip sampling block. This model has been tested on an improved Crack500 dataset. By training and comparing FCN, U-Net and ResU-Net with the same structure, it is concluded that CrackW-Net has the best performance, and its IOU is around 10% higher than U-Net.

(3) CrackW-Net(U) shows similar performance to CrackW-Net (ResU), although the parameter of the former is 1/4 of the latter. It shows that the effect of residual blocks in the task of crack segmentation using CrackW-Net is not ideal. However, due to the simple structure change of CrackW-Net(ResU), the mainstream lightweight methods can be directly applied to reduce the network size.

In future, further validations are still required for CrackW-Net in order to make it practical applicable. Three aspects of validations are suggested for the implementations. The first aspect is to test the performance of CrackW-Net in diverse datasets to determine the boundary of performance improvement. The second aspect should be focused on optimizing the structure of CrackW-Net(ResU) with the purpose of lightweight realization, i.e., to reduce the training parameters of CrackW-Net with residual block without reducing the performance of the network. The third aspect is the optimization of the loss function. Several scholars have put forward a series of loss functions about the pavement crack segmentation, which are worthwhile of further investigating to quantify the amount of improvements that the network segmentation can achieve if they are applied to CrackW-Net.

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Chengjia Han received the B.Sc. degree from the Wuhan University of technology, Wuhan, China. He is currently pursuing the Ph.D. degree with the School of Transportation, Southeast University, Nanjing, China.

His main research interest is deep learning techniques for pavement maintenance and management.

Tao Ma received the B.Sc. degree in civil engineering and the Ph.D. degree in road and railway engineering from Southeast University, Nanjing, China, in 2004 and 2010, respectively.

He is currently a Professor with the School of Transportation, Southeast University, China. He is the Deputy Dean of the School of Transportation and the Head of the Road Engineering Department, Southeast University, the Deputy Director of the National Engineering Laboratory of New Road Materials, the Director of the Key Laboratory of Jiangsu Province of Long-Term Service and Safety of Road Infrastructure, and the Deputy Director of the Road Traffic Engineering National Experimental Teaching Center and the Virtual Simulation of Road Traffic Engineering.

Prof. Ma was selected as a member of the Technical Committee of National Local Joint Engineering Laboratory of Transportation Civil Engineering Materials, the Young Experts Committee of China Highway Society, and the Young Experts Committee of China Highway Journal. He was one of the Leading Talents in Science and Technology Innovation and National Outstanding Youth of the Ministry of Transportation.

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Ju Huyan received the B.Sc. and M.Sc. degrees from Chang'an University, Xi'an, China. She is currently pursuing the Ph.D. degree with the Centre for Pavement and Transportation Technology (CPATT).

In 2021, she joined the School of Transportation, Southeast University, Nanjing, China. Her current research fields include pavement management, pavement condition evaluation, 3D (2D) image processing pavement distress detection, artificial intelligent (AI), machine learning, and deep learning techniques for pavement condition assessment and evaluation.

Xiaoming Huang received the B.Sc. degree in civil engineering and the Ph.D. degree in road and railway engineering from Southeast University, Nanjing, China, in 1985 and 1990, respectively.

He is currently a Professor with the School of Transportation, Southeast University. He was selected as the New Century Academic Leader of Jiangsu Higher Education Institutions, China, the First-Level Candidate for the 110th Talent Project of the New Century of the Ministry of Communications, China, and the Young and Middle-Aged Science and Technology Leader in the 333 High-level Talent Cultivation Project, China.

Yanning Zhang received the B.Sc. degree in mechanical engineering from Southeast University, Nanjing, China, in 2003.

She is currently a Mid-Level Engineer with the School of Transportation, Southeast University, China. Her main research interests include pavement management and road assessment based on image recognition.

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